INFOSYS SPRINGBOARD 6.0

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Topic- Smart Energy Consumption Analysis and Prediction using Machine Learning with Device-Level Insights

Documentation for Week 3-4

```python
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import numpy as np

# Initialize model
lr_model = LinearRegression()

# Train model
lr_model.fit(X_train, y_train)

print("Linear Regression model trained successfully ")
```

```
Linear Regression model trained successfully
```

Double-click (or enter) to edit

```python
# Predictions
y_train_pred = lr_model.predict(X_train)
y_val_pred = lr_model.predict(X_val)
y_test_pred = lr_model.predict(X_test)

print("Predictions completed ")
```

```
Predictions completed
```

A Linear Regression model was implemented as the baseline forecasting model to predict energy consumption. The model was trained using the training dataset and used to generate predictions for the training, validation, and test sets. This baseline approach establishes a reference performance level, allowing evaluation of prediction accuracy using metrics such as MAE, RMSE, and $R^2$, and serves as a benchmark for comparison with more advanced models like Random Forest.

```python
# Evaluation on Test Set

lr_mae = mean_absolute_error(y_test, y_test_pred)
lr_rmse = np.sqrt(mean_squared_error(y_test, y_test_pred))
lr_r2 = r2_score(y_test, y_test_pred)

print("Baseline Model Performance (Test Set)")
print("MAE :", round(lr_mae, 4))
print("RMSE:", round(lr_rmse, 4))
print("R2  :", round(lr_r2, 4))
```

```
Baseline Model Performance (Test Set)
MAE : 0.4104
RMSE: 0.4904
R2  : 0.6639
```

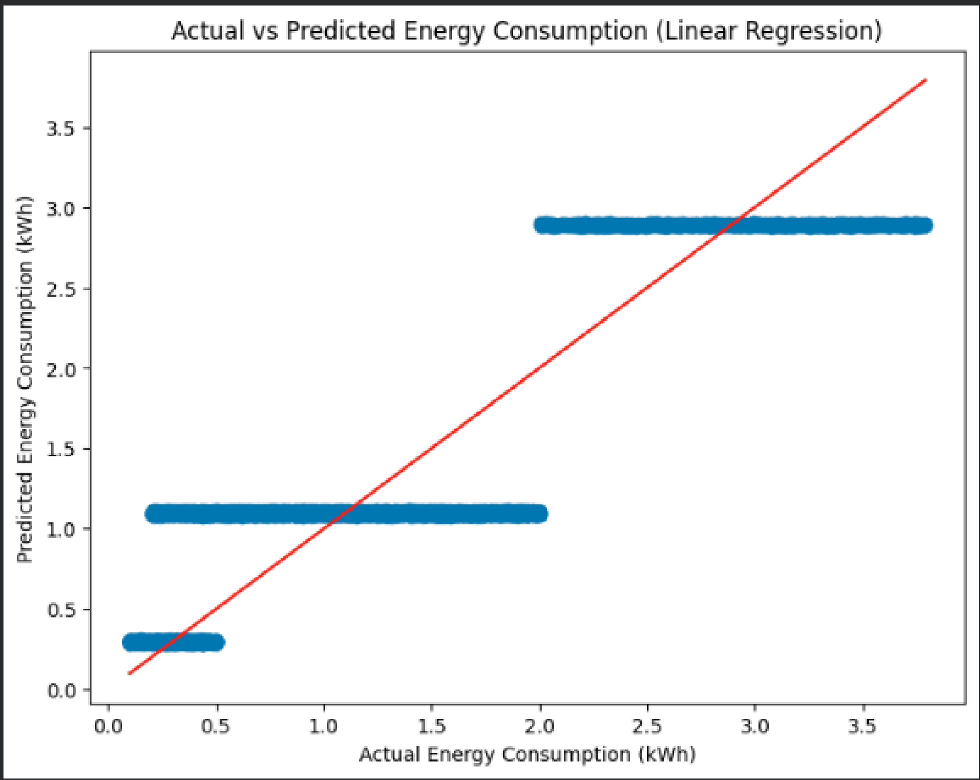
The Linear Regression baseline model achieved an MAE of 0.4104 and RMSE of 0.4904, indicating moderate prediction error.
With an $R^2$ score of 0.6639, the model explains approximately 66% of the variance in energy consumption.

```python
plt.figure(figsize=(8,6))

plt.scatter(y_test, y_test_pred, alpha=0.4)
plt.xlabel("Actual Energy Consumption (kWh)")
plt.ylabel("Predicted Energy Consumption (kWh)")
plt.title("Actual vs Predicted Energy Consumption (Linear Regression)")

plt.plot([y_test.min(), y_test.max()],
         [y_test.min(), y_test.max()],
         color='red')

plt.show()
```




The scatter plot shows that predictions are clustered in horizontal bands rather than closely following the 45-degree reference line. This indicates that the Linear Regression model captures general trends but struggles to model finer variations in energy consumption.

```python
coefficients = pd.DataFrame({
    "Feature": X_train.columns,
    "Coefficient": lr_model.coef_
}).sort_values(by="Coefficient", ascending=False)

coefficients.head(10)
```

| | Feature | Coefficient |
|---|---|---|
| 4 | Day | 0.000028 |
| 0 | Home ID | -0.000010 |
| 1 | Outdoor Temperature (ï¿½ï | -0.000082 |
| 3 | Hour | -0.000211 |
| 6 | Weekday | -0.000568 |
| 16 | Season_Spring | -0.001122 |
| 5 | Month | -0.001488 |
| 10 | Appliance Type_Heater | -0.001966 |
| 2 | Household Size | -0.003432 |
| 17 | Season_Summer | -0.006343 |

The coefficient analysis shows that most features have very small coefficient values, indicating limited individual impact on energy consumption in the linear model. No single feature strongly dominates the prediction, suggesting that energy usage is influenced by multiple factors collectively rather than one primary variable.

```python
from sklearn.ensemble import RandomForestRegressor

rf_model = RandomForestRegressor(
    n_estimators=100,
    random_state=42
)

rf_model.fit(X_train, y_train)

y_pred_rf = rf_model.predict(X_test)

# Metrics
rf_mae = mean_absolute_error(y_test, y_pred_rf)
rf_rmse = np.sqrt(mean_squared_error(y_test, y_pred_rf))
rf_r2 = r2_score(y_test, y_pred_rf)
```

```python
comparison = pd.DataFrame({
    "Model": ["Linear Regression", "Random Forest"],
    "MAE": [lr_mae, rf_mae],
    "RMSE": [lr_rmse, rf_rmse],
    "R2 Score": [lr_r2, rf_r2]
})

comparison
```

| | Model | MAE | RMSE | R2 Score |
|---|---|---|---|---|
| 0 | Linear Regression | 0.410420 | 0.49036 | 0.663869 |
| 1 | Random Forest | 0.417403 | 0.50310 | 0.646176 |

The comparison between Linear Regression and Random Forest shows that Linear Regression slightly outperformed Random Forest in terms of MAE, RMSE, and $R^2$ score. This suggests that the relationship between the features and energy consumption is largely linear in nature. Both models achieved moderate predictive performance, explaining approximately 65% of the variance in energy consumption.

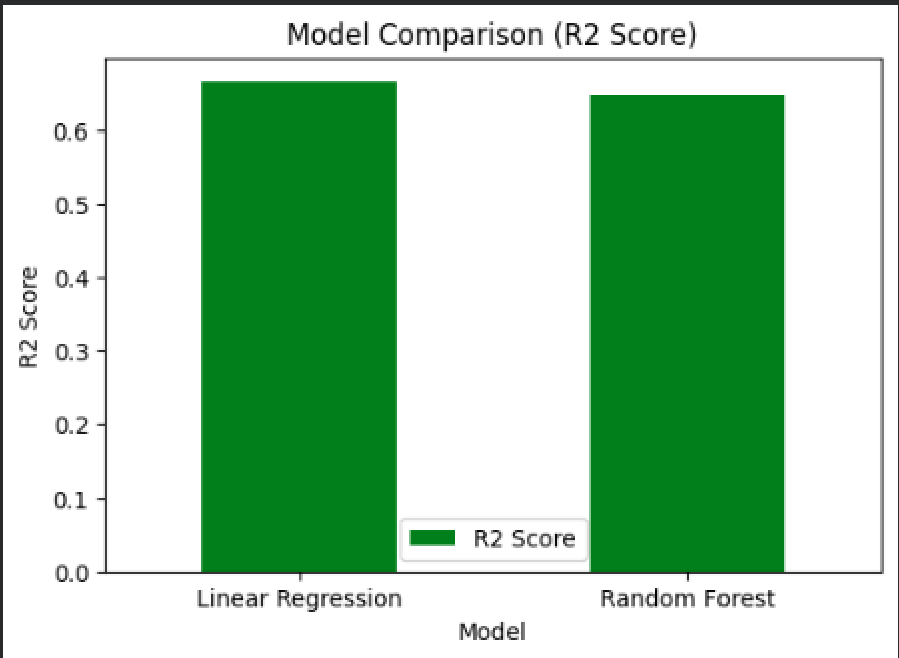
```python
comparison.set_index("Model")[["MAE", "RMSE"]].plot(
    kind='bar',
    figsize=(6,4)
)

plt.title("Model Comparison (Error Metrics)")
plt.ylabel("Error Value")
plt.xticks(rotation=0)
plt.show()
```


The error comparison chart shows that Linear Regression has slightly lower MAE and RMSE values compared to Random Forest. This indicates that the Linear Regression model performs marginally better in terms of prediction accuracy on the test set, suggesting that the underlying relationship in the dataset is largely linear.

```
comparison.set_index("Model")[["R2 Score"]].plot(
    kind='bar',
    figsize=(6,4),
    color='green'
)

plt.title("Model Comparison (R2 Score)")
plt.ylabel("R2 Score")
plt.xticks(rotation=0)
plt.show()
```




The $R^2$ comparison chart shows that Linear Regression achieves a slightly higher $R^2$ score than Random Forest, explaining about 66% of the variance in energy consumption. This indicates that the dataset follows a predominantly linear pattern, allowing the simpler linear model to perform marginally better than the more complex ensemble model.